CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

Bachelor of Technology(B.Tech) (CL/EC/ME/ CE/EE/IT)

Semester- I University Examination May 2010

MA101 Engineering Mathematics-I

Date:25/5/2010 , Tuesday Time: 12:30 p. m to 3:30 p. m. Max. Marks:70

Instructions:

1. **Attempt all the questions in each section.**
2. **Assume suitable additional data if required.**
3. **Figures to the right side indicate the full marks.**

SECTION-I

Q-1 (a) Determine the rank of the matrix **[3]**

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

(b) Prove that **[4]**

$$\frac{(\cos 2\theta - i\sin 2\theta)^5 (\cos 3\theta + i\sin 3\theta)^6}{(\cos 4\theta + i\sin 4\theta)^7 (\cos\theta - i\sin\theta)^8} = \cos 12\theta - i\sin 12\theta$$

Q-2 (a) Find the Rank and Nullity of **[5]**

$$A = \begin{bmatrix} 3 & 4 & 5 & 6 & 7 \\ 4 & 5 & 6 & 7 & 8 \\ 5 & 6 & 7 & 8 & 9 \\ 10 & 11 & 12 & 13 & 14 \\ 15 & 16 & 17 & 18 & 19 \end{bmatrix}.$$

(b) Examine for consistency & if consistent solve them **[5]**

$x + 2y = 3,\ y - z = 2,\ x + y + z = 1.$

(c) Find the root of the equation $z^4 + 1 = 0$. **[5]**

(d) Using D'Moivre's theorem prove that, **[5]**

$\cos 6\theta = 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1.$

OR

Q-2 (a) By using Gauss Jordon reduction method find the inverse of A **[5]**

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}.$$

(b) Examine whether the following equations are consistent and solve if consistent $2x + 6y + 11 = 0,\ 6x + 20y - 6z + 3 = 0,\ 6y - 18z + 1 = 0.$ **[5]**

(c) Expand $\sin^5\theta\cos^3\theta$ in a series of sine multiples of θ. **[5]**

(d) If $x+\frac{1}{x}=2\cos\theta,\ y+\frac{1}{y}=2\cos\varphi\ \&\ z+\frac{1}{z}=2\cos\psi$, then prove that [5]

$$x^p y^q z^r+\frac{1}{x^p y^q z^r}=2\cos\left(p\theta+q\varphi+r\psi\right).$$

Q-3 Test the convergence of the following series (Any two) [8]

$$(i)\ \sum_{n=0}^{\infty}\frac{2^{3n}}{3^{2n}},\quad (ii)\ \sum_{n=0}^{\infty}\frac{\tan^{-1}n}{n^2+1},\quad (iii)\ \sum_{n=0}^{\infty}\frac{e^n}{n!}$$

OR

Q-3 Test the convergence of the following series (Any two) [8]

$$(i)\sum_{n=0}^{\infty}\frac{x^n}{2n+1},\quad (ii)\sum_{n=0}^{\infty}\frac{x^n}{n\ln n},\quad (iii)\ \sum_{n=0}^{\infty}3^n x^{2n}$$

SECTION-II

Q-4 (i) Find the nth derivative of $y=\sin x\cos x$ [3]

(ii) Evaluate: $\lim_{x\to\pi/2}\left(\sec x-\tan x\right)$ [2]

Q-5 (i) Express $\sin(a+h)$ as a series of powers of h and hence evaluate the value of $\sin 62^\circ$ correct up to four decimal places. [5]

(ii) State Roll's Theorem. Verify Rolle's theorem for $f(x)=x^2$ in $-1\le x\le 1$. [5]

(iii) Use Lagrange's Mean Value Theorem, find 'C' if [5]

$$f(x)=x(x-1)(x-2)\ \text{ in }\ \left[0,\frac{1}{2}\right].$$

OR

Q-5 [5]

(i)Find the Maclaurin's series of $\sin\left(\frac{\pi}{3}+x\right)$.

(ii) State Cauchy's Mean Value Theorem. Verify it for $f(x)=\sin x$ and $g(x)=\cos x$ and show that 'c' of the Cauchy's Mean value theorem is the average of $'a'$ and $'b'$. [5]

(iii)Verify Rolle's theorem for $(x+1)^m(x-1)^n\ in -1\le x\le 1$ where m,n are positive integers. [5]

Q-6 (i) Using Euler's Theorem on Homogeneous functions, prove that **[5]**

If $u = \tan^{-1}\left(\frac{x^3 + y^3}{x - y}\right)$, then $x\,u_x + y\,u_y = \sin 2u$.

(ii) Find the equation of tangent plane and the normal line to the surface **[5]**
$2x^2 + y^2 + 2z = 3$ at the point $(2,1,-3)$.

(iii) Find the minimum of $f(x,y) = xy + 27\left(\frac{1}{x} + \frac{1}{y}\right)$. **[5]**

OR

Q-6 (i) Find the percentage error in computing the resistance r from the formula $\frac{1}{r_1} = \frac{1}{r_2} + \frac{1}{r}$, if r_1, r_2 are both in error by 2%. **[5]**

(ii) Find the maximum value of $f(x,y,z) = xyz$, subject to the constraint $2x + 2y + 2z = 108$ **[5]**

(iii) If $u = \frac{y^2}{x}, v = \frac{x^2}{y}$, find $J = \frac{\partial(x,y)}{\partial(u,v)}$ and $J' = \frac{\partial(u,v)}{\partial(x,y)}$ and hence verify that $J J' = 1$. **[5]**
